

Suraj Shukla

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AI Engineer • Python, Rust, TypeScript • LLM Agents, RAG, Multi-Agent Orchestration

PROFESSIONAL EXPERIENCE

Divergent Classes, Software Engineer Delhi, India	2026 to present
- Built the core CRM/LMS managing course enrollment, per-lecture progress, and test results across the student base, with rule-based pacing personalization. - Built NIFT/UCEED college-admission predictors that estimate chances from student score and rank against historical cutoffs. - Built a practice-problem platform with past-year question banks, daily practice sessions, and per-student accuracy and improvement tracking over time.	
Bramer Private Limited, Co-founder & CTO Delhi, India	2022 to 2026
- Led a cross-functional team of 6 across software, AI, electronics, and hardware on a \$300K (INR 3 crore) R&D program. - Built cross-platform ground control station, an event-driven backend, and CUDA-accelerated edge inference on Jetson AGX, converting YOLOv8 detection and custom segmentation models to TensorRT engines for real-time onboard processing under a fixed power and thermal budget. - Built a Redis-backed job durability layer for the analysis pipeline: worker liveness and per-job progress persisted externally, so a dead worker's in-progress job is picked up and resumed by another worker without re-running completed steps.	

PROJECTS

fabric Rust, Tokio, Axum, LangGraph, TypeScript, React, MAVLink, Tauri 2, CUDA, TensorRT, Kubernetes	app.bramer.in
- Built a LangGraph agent that takes a goal like "survey this area" and plans which drones, AUVs, or UGVs do what, and when, across the whole fleet. - Rust hub-core reaches a separate Python LangGraph agent service over HTTP and gRPC, so the agent's planning stays isolated from the flight-critical control path. - The agent only sees each asset's real commands as typed tool schemas, so it emits a constrained action plan, not free text, and can't invent a command a vehicle can't run. - Human in the Loop: No plan reaches a vehicle without a human approving it first. - Safety checks (geofence, arm state) run in deterministic code outside the agent, with every decision logged and a kill switch that halts execution mid-flight.	
Levr Next.js 16, Vercel AI SDK, LangGraph, FastAPI, PostgreSQL, MCP, Meta Ads API	levr200ok.vercel.app github.com/beingglitch/levr
- Plans and runs ad campaigns from a stated goal and budget, using a LangGraph agent driven by structured tool calls. - Built an MCP server exposing the Meta Ads API as tools, so any MCP-compatible client, not just this app's own agent, can read and act on ad account data through a standard protocol. - Built the agent to reason in platform-agnostic primitives (audience, budget, creative) instead of Meta's specific API, so adding Google or X later is a new adapter. - Nothing spends money without human approval first, with every tool call schema-validated before it reaches a live ad platform.	
Nirman Agentic Construction ERP. Next.js 16, React 19, Prisma 7, PostgreSQL, LangChain, LangGraph	nirmanos.in
- Embeds site reports, contracts, and incident logs into a vector store and answers "what's going wrong and what should we do" with a RAG pipeline that cites the actual record behind each claim. - Replaced 10+ departments' manual tracking with one shared system, so the agent retrieves against today's data, not last week's report.	
Oxide Rust	crates.io/oxide-cipher github.com/beingglitch/oxide
Built a file encryption CLI from scratch with no cryptography crates, implementing the cipher, key derivation, and streaming architecture at the byte level.	
Lookup Go - goroutines, context, channels	github.com/beingglitch/lookup
- Built a recursive DNS resolver from scratch, including binary protocol encoding/decoding over raw UDP, with no standard or third-party library used. - Queried multiple nameservers in parallel at each level of the delegation chain and raced them for the fastest response, using goroutines, channels, and context-based timeouts. - Distributed as both an importable Go module and a CLI package via go install.	

SKILLS

AI and Agentic Systems: LangGraph, LangChain, Vercel AI SDK, MCP (Model Context Protocol), RAG, tool-calling and structured output, multi-agent orchestration
GPU and Edge Inference: CUDA, TensorRT, NVIDIA Jetson, PyTorch, OpenCV, YOLOv8
Languages: Python, Rust, Go, TypeScript/JS, C++, SQL
Systems and Distributed: Control planes, state machines and reconciliation loops, event-driven architecture, NATS, gRPC, REST, WebSocket, idempotency and retry design
Backend: Tokio, Axum, FastAPI, Node.js
Data: PostgreSQL, PostGIS, Redis, Cassandra, MongoDB, SQLite, Prisma
Infrastructure: Kubernetes, Docker, Linux, Nginx, GitHub Actions CI/CD, AWS, GCP
Frontend: React, Next.js, React Native, Tauri 2, TanStack, Zustand, TailwindCSS
Robotics: ROS, Gazebo, MAVLink

LEADERSHIP & RESEARCH ROLES

I-Hub Foundation for Cobotics (IHFC), IIT Delhi, Research Fellow	2026 to present
Selected for a competitive fellowship in collaborative robotics at IHFC, a Department of Science and Technology Innovation Hub at IIT Delhi to build a swarm-coordination system for multi-agent robotics.	
DTU IoT Research Lab, Head of Development	2025 to present
Leading a team of student developers across different research & development projects in computer vision, signal processing, software defined networks (SDN) & robotics applications.	
DTU Unmanned Aerial Systems (UAS) Team, Software & Autonomy Lead	2022 to 2024
Led the software and autonomy sub-team; developed a swarm-coordination framework for drones with GPS-denied visual navigation & action pipeline on ROS 2.	

ONGOING RESEARCH

Neuromorphic Simultaneous Localization And Mapping (SLAM): Developing a Spiking Neural Network (SNN) based SLAM algorithm for low-energy, vision-only navigation on micro-swarm drones.
Subsurface Mapping: Ground-Penetrating Radar (GPR) signal processing and machine-learning segmentation of B-scan and C-scan data to reconstruct subsurface structures for non-invasive mapping.

EDUCATION

Delhi Technological University (DTU) B.Tech in Software Engineering	Delhi, India
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